TASK 03:

```
//Tasks 3: You are tasked with developing a program to manage and display the Grade Point Average
//(GPA) for the core courses offered in the first semester of four departments: Software Engineering (SE),
//Artificial Intelligence (AI), Computer Science (CS), and Data Science (DS). Each department offers a
//distinct number of core courses for this semester: SE has 3 core courses, AI has 4 core courses, CS has 2
//core courses, and DS has 1 core course. To efficiently store and present this data, which type of array
//structure would you employ? implement a solution using the chosen array structure to display the GPAs
//of the core courses for each department.

#include<iostream>
using namespace std;

int main()
{
  int dept=4;
  int **arr=new int *[dept];
  int size[4];

  for(int i=0;i<dept;i++)
  {
    cout<<"\nEnter size of dept "<<i+1<<": ";
    cin>>size[i];

    arr[i]=new int[size[i]];
  }

  for(int i=0;i<dept;i++)
  {
    cout<<"\nEnter gpa of "<< size[i] <<" core courses for dept "<<i+1<<": ";
    for(int j=0;j<size[i];j++)
    {
      cout<<"\nCore Course "<<j+1<<" : ";
      cin>>arr[i][j];
    }
  }

  cout<<"\n\nThe entered data is : ";
```

```
    for(int i=0;i<dept;i++)
    {
        cout<<"\nDept "<<i+1<<": ";

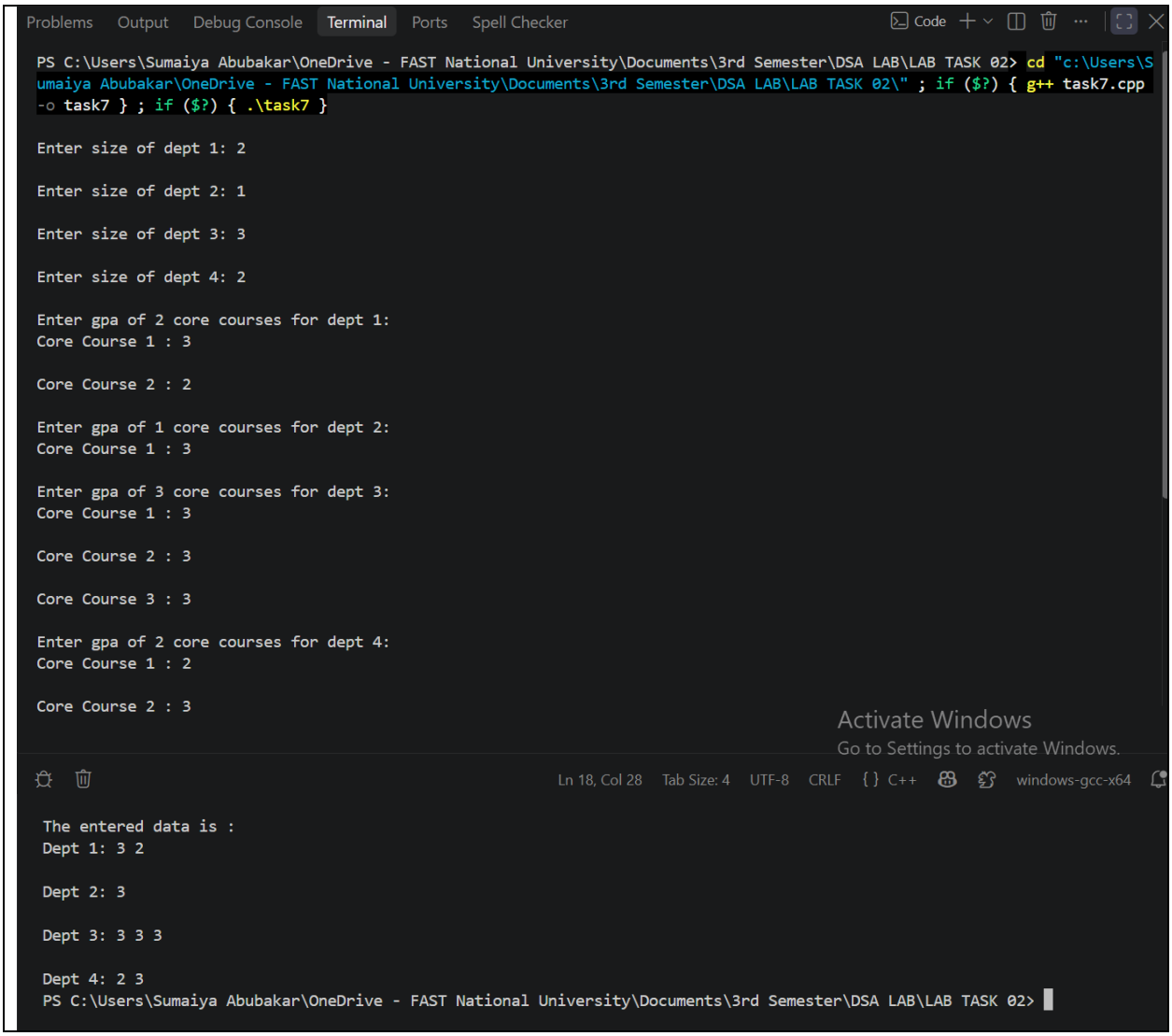
        for(int j=0;j<size[i];j++)
        {
            cout<<arr[i][j]<<" ";
        }

        cout<<endl;
    }

    for(int i=0;i<dept;i++)
    {
        delete[] arr[i];
    }

    delete[] arr;

    return 0;
}
```



TASK 04:

```
#include<iostream>
using namespace std;

int main()
{
   int rows;

   cout<<"Enter number of rows: ";
   cin>>rows;

   string** seatingChart=new string*[rows];
```

```
int* seatsPerRow=new int[rows];

for(int i=0;i<rows;i++)
{
  cout<<"\nEnter number of seats in row "<<i+1<<" : ";
  cin>>seatsPerRow[i];

  seatingChart[i]=new string[seatsPerRow[i]];
  cin.ignore();
  for(int j=0;j<seatsPerRow[i];j++)
  {
    cout<<"\nEnter name for seat "<<j+1<<" : ";
    // cin>>seatingChart[i][j];

    getline(cin,seatingChart[i][j]);
  }
}

cout<<"\n\nDISPLAYING SEATING CHART\n\n";

for(int i=0;i<rows;i++)
{
  cout<<"Row "<<i+1<<" : "<<endl;

  for(int j=0;j<seatsPerRow[i];j++)
  {
    cout<<seatingChart[i][j]<<" ";
  }

  cout<<endl;
}

for(int i=0;i<rows;i++)
{
  delete[] seatingChart[i];
}

delete[] seatingChart;
delete[] seatsPerRow;

seatingChart=NULL;
seatsPerRow=NULL;

return 0;
```

```
}
```

```
Problems  Output  Debug Console  Terminal  Ports  Spell Checker                Code

umaiya Abubakar\OneDrive - FAST National University\Documents\3rd Semester\DSA LAB\LAB TASK 02\" ; if ($?) { g++ task8.cpp
-o task8 } ; if ($?) { .\task8 }
Enter number of rows: 3

Enter number of seats in row 1 : 2

Enter name for seat 1 : Sumaiya

Enter name for seat 2 : Aliza

Enter number of seats in row 2 : 3

Enter name for seat 1 : Roha

Enter name for seat 2 : Hafsa

Enter name for seat 3 : Hania

Enter number of seats in row 3 : 1

Enter name for seat 1 : Jenny


DISPLAYING SEATING CHART

Row 1 :
Sumaiya Aliza
Row 2 :
Roha Hafsa Hania
Row 3 :
Jenny
                                                        Activate Windows
```

TASK 05:

```
#include<iostream>
using namespace std;

int main()
{
   int rows;

   cout<<"Enter number of rows: ";
   cin>>rows;

   int** elements=new int*[rows];

   int* elementsPerRow=new int[rows];

   for(int i=0;i<rows;i++)
   {
      cout<<"\nEnter number of elements of row "<<i+1<<" : ";
```

```
    cin>>elementsPerRow[i];

    elements[i]=new int[elementsPerRow[i]];

    for(int j=0;j<elementsPerRow[i];j++)
    {
        cout<<"\nElement "<<j+1<<" : ";
        cin>>elements[i][j];
    }
}
int maxElements=0;
int maxRow=0;
for(int i=0;i<rows;i++)
{
    if(elementsPerRow[i]>maxElements)
    {
        maxElements=elementsPerRow[i];
        maxRow=i;
    }
}
cout << "\nThe row with maximum elements is row " << maxRow + 1 << endl;
cout << "Size of that row: " << maxElements << endl;

cout << "Elements of that row: ";

for(int j=0; j<elementsPerRow[maxRow]; j++)
{
cout << elements[maxRow][j] << " ";
}

for(int i=0; i<rows; i++)
{
    delete[] elements[i];
}

delete[] elements;
delete[] elementsPerRow;

elements=NULL;
elementsPerRow=NULL;

return 0;
```

```
}
```

```
Enter number of rows: 3

Enter number of elements of row 1 : 2

Element 1 : 10

Element 2 : 20

Enter number of elements of row 2 : 1

Element 1 : 89

Enter number of elements of row 3 : 3

Element 1 : 45

Element 2 : 67

Element 3 : 81

The row with maximum elements is row 3
Size of that row: 3
Elements of that row: 45 67 81
PS C:\Users\Sumaiya Abubakar\OneDrive - FAST National University\Documents\3rd Semester\DSA LAB\LAB TASK 02>
```